

MERN Stack Fresher/Intern Role (1 Hour)

Instructions:

- Total Time: 1 hour
 - Attempt all questions
 - Write your code using **Node.js + Express (backend)** and **React (frontend)**
 - Use **MongoDB** for database (local or Atlas)
 - Ensure clean, well-documented code following best practices
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Section 1: Backend Development (Node.js + Express + MongoDB)

Question 1: RESTful API Development (30 minutes)

Create a simple RESTful API using **Express.js** to manage a product catalog for a fictional eCommerce store ("FakeStore").

Reference API: <https://fakestoreapi.com/products>

Requirements:

Implement the following endpoints:

- **GET /products**
Retrieve a list of all products from MongoDB
- **POST /products**
Add a new product
Request body should include:
 - name
 - description
 - price
 - image_url
 - category
- **PUT /products/:id**
Update an existing product's details by ID
- **DELETE /products/:id**
Delete a product by its ID

- **POST /cart**
Add a product to the cart (store cart data in MongoDB or in-memory)
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Technical Expectations:

- Use **Express Router**
 - Use **MongoDB with Mongoose**
 - Proper schema design for Product
 - Input validation & error handling
 - Use **async/await**
 - Follow RESTful conventions
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Section 2: Frontend Development (React)

Question 2: Frontend Application (20 minutes)

Create a simple **React application** that interacts with the backend API developed in Question 1.

Requirements:

- Display a list of products fetched from **GET /products**
 - Provide a form to add a new product (POST /products)
 - Implement a button to add products to the cart (POST /cart)
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Technical Expectations:

- Use **functional components + hooks (useState, useEffect)**
- Use **Axios or Fetch API** for API calls
- Basic styling (CSS / Tailwind optional)
- Proper state management